Paleomagnetism of the Proterozoic Wathaman batholith and the suturing of the Trans-Hudson orogen in Saskatchewan[1]

D. T. A. SYMONS
Department of Geology, University of Windsor, Windsor, Ont., Canada N9B 3P4

Received April 3, 1991
Revision accepted June 27, 1991

The Wathaman (Wathaman–Chipewyan) batholith is an 1854 ± 11 Ma, northeast-trending, homogenous, felsic pluton that is over 900 km long. It is thought to be a magmatic arc, with the Archean Hearne craton on its northwestern side as the hinterland and the remains of the Early Proterozoic Manikewan oceanic crust of the Trans-Hudson orogen on its southeastern side. Alternating-field and thermal step demagnetization methods isolate an A remanence component with a mean direction of D = 134.6°, I = 54.1° (α_{95} = 3.5°, k = 94, N = 19). Isothermal remanent magnetization tests confirm that this A magnetization component is preserved in pseudosingle to multidomain magnetite and in hematite. Contact tests with intruded older rocks of the Peter Lake domain, with younger crosscutting mafic dikes and with younger crosscutting shear zones, indicate that A is a primary remanence. Its pole position of 67°W, 9°N (dp = 3°, dm = 5°) confirms that it was formed along the margin of the Slave–Rae–Hearne craton, supporting tectonic models that it records a suture zone. It also indicates that the LaRonge – Lynn Lake domain, Flin Flon domain, and Superior Province were translated relatively northwestward into the suture by at least 11 ± 11°, 27 ± 12°, and 49° ± 16°, respectively, as the Manikewan Ocean closed.

Le batholite de Wathaman (Wathaman–Chipewyan) est un pluton felsitique, homogène, âgé de 1854 ± 11 Ma, de direction nord-est et d'une longueur de plus de 900 km. Il est interprété comme un arc magmatique côtoyé sur le nord-ouest par le craton Hearne archéen représentant l'arrière-pays, et sur le sud-est par les vestiges de la croûte océanique protérozoïque précoce de l'océan Manikewan de l'orogène Trans-Hudson (OTH). La désaimantation par les méthodes de champ alternatif et de chauffage isole une composante de rémanence A avec une direction moyenne de D = 134,6°, I = 54,1° (α_{95} = 3,5°, K = 94, N = 19). Les tests d'aimantation rémanente isotherme confirment que la composante A d'aimantation est préservée dans les pseudomonodomaines et polydomaines de la magnétite et de l'hématite. Les tests du contact avec les roches intrusives plus anciennes du domaine structural de Peter Lake, avec les dykes mafiques sécants plus jeunes, et avec les zones de cisaillement transversales plus jeunes révèlent que A représente une rémanence primaire. La position de son pôle de 67°W, 9°N (dp = 3°, dm = 5°) confirme qu'elle fut acquise le long de la marge du craton de Slave–Rae–Hearne, étayant les modèles tectoniques qui invoquent une zone de suture. En plus, ceci indique que le domaine de LaRonge – Lynn Lake, le domaine de Flin Flon, et la province du lac Supérior ont dérivé relativement vers le nord-ouest dans la zone de suture de 11 ± 11°, 27 ± 12°, et 49 ± 16° au moins, respectivement, lors de la fermeture de l'océan Manikewan.

[Traduit par la rédaction]

Can. J. Earth Sci. **28**, 1931–1938 (1991)

Introduction

Recently Dunsmore and Symons (1990) showed that the available paleomagnetic poles from 1825 to 1900 Ma for the Archean Slave and Superior cratons and intervening Early Proterozoic Trans-Hudson orogen (THO) fit remarkably well to a collisional closure geotectonic model. They interpreted the data to show that the Slave craton was translated about 13° southeast and the Superior craton about 30° northwest into the THO. In the process the Manikewan Ocean basin (Stauffer 1984), with a minimum width of about 4000 km, was closed. The Wathaman batholith is thought to be an ensialic magmatic arc that defines the location of the northwest-dipping THO suture zone. Thus it is an obvious target for paleomagnetic study to better understand the geometry of closure of the THO.

Geology

The Wathaman batholith, sometimes called the Wathaman–Chipewyan batholith, in Saskatchewan and Manitoba is a megacrystic granitoid pluton that is over 900 km long and over 150 km wide in places, making it the world's largest known Precambrian pluton (Fig. 1) (Fumerton *et al.* 1984). This massive monzogranite–granodiorite pluton is remarkably homogeneous and shows little evidence of multiple intrusive episodes (Lewry *et al.* 1981). The batholith's northwestern contact is defined by blastomylonites in the major Needle Falls and Parker Lake shear zones that separate it from the Wollaston and Seal River fold belts and the Archean Peter Lake domain. The fold belts define the deformed margin of the Archean Hearne province, outboard of which are the Rae and Slave provinces (Hoffman 1988). On the southeastern side of the Wathaman batholith is the intervening juvenile Early Proterozoic crust of the THO. This zone is composed of intraoceanic rocks of several domains which are relicts of the Manikewan oceanic basin and extend southeastwards to the Archean Superior province (Stauffer 1984).

The Wathaman batholith is comprised of richly porphyritic quartz monzonite, grading to granodiorite and granite in this study area (Lewry *et al.* 1980*a*, 1980*b*; Johnston 1985). The abundance and size of the microcline megacrysts decrease near the batholith's margins. A weak primary flow foliation, acquired during magmatic emplacement, is seen in the preferential alignment of the megacrysts. Superimposed on the primary fabric is the main flaser and mylonite fabric associated with the shearing and compressional event that formed the Needle Falls and Parker Lake shear zones. This event produced blastomylonites and submylonites to flasered augen gneisses in a 0.5–1.5 km wide zone of sheared granite next to the northwestern contact. The intensity of deformation decreases inwards, so only localized splaying shear zones transect most of the batholith (Green *et al.* 1985*a*). Rare metapyroxenite bands are present that parallel the north east-trending, steeply northwest-dipping (>70°) foliation. They resemble altered

[1] Lithoprobe Publication 243.

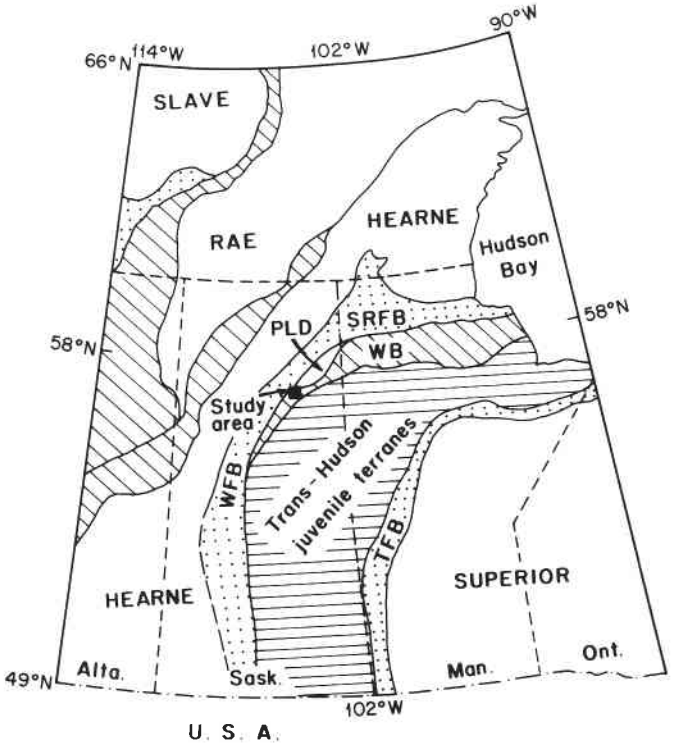


FIG. 1. Location of the study area, with the main tectonic elements from Hoffman (1989) including Peter Lake domain (PLD), Seal River fold belt (SRFB), Thompson fold belt (TFB), Wathaman batholithic arc (WB), and Wollaston fold belt (WFB).

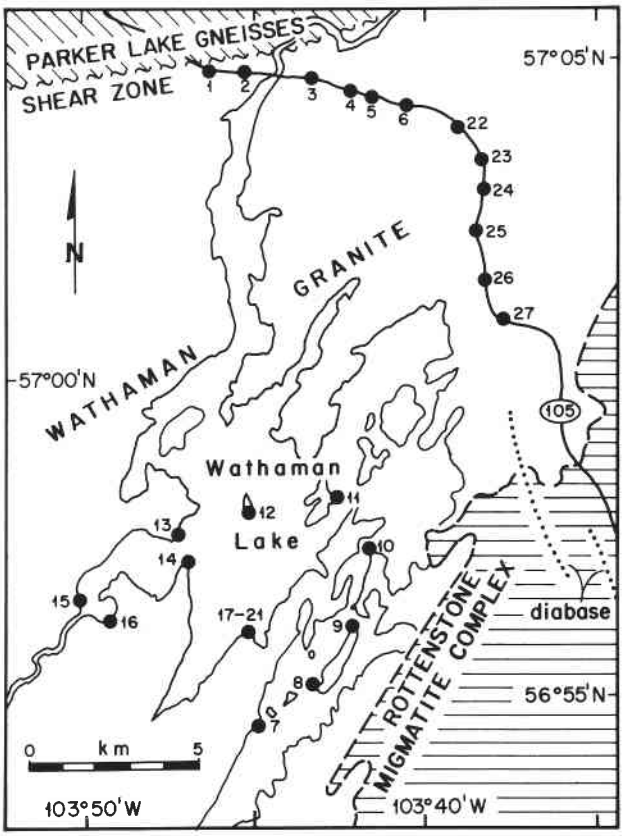


FIG. 2. Geology of the study area after Lewry *et al.* (1980*a*, 1980*b*) and Johnston (1985), with site locations.

mafic dikes but could be inclusions (Johnston 1985). Late screens of aplite or red–pink pegmatite cut the batholith and terminate the emplacement event. The batholith has yielded U–Pb zircon dates, mostly from in or near the study area, of 1870 ± 20, 1862 ± 4, 1856 ± 8, 1852 ± 14, 1852 ± 6, and 1830 ± 6 Ma (Ray and Wanless 1980; Meyer 1987; Van Schmus *et al.* 1987), with a mean of 1854 ± 11 Ma. A very few narrow unmetamorphosed gabbro–diabase dikes cut the batholith along a north-northwest trend. The batholith is also overlain by an outlier of flat-lying unmetamorphosed quartz-pebble conglomerate of the Athabaska Formation just east of the study area (Lewry 1980*b*). Deposition of this formation began between about 1720 Ma and 1750 (Hoffman 1989) as the Trans-Hudson orogeny terminated (Machado 1990).

Experimental methods

The Wathaman batholith was sampled at 27 sites along Highway 105 and the shoreline of Wathaman Lake (Fig. 2). Of these, 24 sites were in granitic rocks and three in narrow foliated dikes for a contact test. Generally five cores were drilled at each site and oriented by sun compass, and two specimens were sliced from each core to give a total collection of 232 specimens (Table 1). Their natural remanent magnetization (NRM) was measured using an automated CTF DRM-420 cryogenic magnetometer inside a shielded room. One specimen from each site was alternating-field (AF) demagnetized in 12 steps up to 120 mT using a Sapphire Instruments SI-4 unit, and a second specimen was thermally demagnetized in 14 steps to 680°C using a Schonstedt TSD-1 unit. The remaining specimens were AF and (or) thermally demagnetized in four to six steps, mostly in either 20 mT increments above 20 mT or in 20°C increments above 500°C to define the characteristic remanence direction of each specimen. As noted below, the granitic rocks behave simply, and uniformly on demagnetization. Accepted characteristic magnetization directions have maximum angular deviation (MAD) values of less than 10° and mostly of less than 3° (Kirschvink 1980). These were averaged to get the site mean directions (Fisher 1953) except in two sites for which remagnetization circle directions were also used (Bailey and Halls 1984). Isothermal remanent magnetization (IRM) tests were run on 10 specimens by magnetizing them in 11 steps up to 900 mT using a Sapphire Instruments SI-6 pulse magnetizer and then AF demagnetizing them in six steps to 50 mT.

Results

The characteristic site mean directions form three obvious groups or magnetization components that reflect geologic controls. The A component is found in massive granite sites and is directed to the southeast with intermediate down inclinations. The B component is found in the northeast-trending mafic dikes and is directed to the northwest with shallow upward inclinations, and the C component in sheared granite is directed steeply downwards. Each component is discussed in turn.

The massive granite sites in the Wathaman batholith with the A component have NRM intensities that are mostly in the low 10^{-1} A/m range with a spread from mid 10^{-3} A/m to mid 10^{0} A/m. AF step demagnetization first removes a low-coercivity viscous remanent magnetization (VRM) component of apparent Holocene origin by 15–20 mT and thereafter isolates the A component over a broad coercivity spectrum up to 100–120 mT (Figs. 3*a*, 3*b*). VRM is typically removed by

TABLE 1. Site mean remanence statistics

Site	No. of specimens (e, r, o)[a]	Demagnetization range used		Mean direction				
		mT	°C	D (°)	I (°)	α_{95} (°)	k	M
1	7, 0, 0	20–80	400–500	210	83	9	47	C
2	4, 0, 1	20–120	450–580	342	79	8	132	C
3	7, 0, 3	10–30	300–450	85	52	13	22	C
4	5, 0, 0	15–30	300–450	125	61	10	62	A
	3, 0, 0		300–450	274	67	25	25	C
5	11, 0, 0	20–60	0–580	135	61	7	48	A
6	11, 0, 0	15–80	500–565	126	64	6	59	A
7	10, 0, 0	5–80	450–580	135	53	5	92	A
8	11, 0, 0	20–80	450–565	140	52	6	50	A
9	11, 0, 0	0–100	0–580	128	42	5	96	A
10	11, 0, 0	10–100	200–580	143	46	7	38	A
11	6, 5, 0	15–100	400–580	141	50	6	66	A
12	8, 0, 0	15–60	400–580	127	56	5	138	A
13	6, 0, 1	15–60	200–565	111	62	12	33	A
14	4, 3, 2	20–80	450–580	138	59	11	39	A
15	11, 0, 0	10–60	530–580	166	75	8	30	C
16	8, 0, 2	15–40	400–580	303	81	11	25	C
17	3, 0, 3	15–60	300	307	−43	23	30	B
18	7, 0, 0	20–100		323	−33	12	33	B
19	2, 0, 4	100–120	300	335	−24		2716	B
20	6, 0, 0	20–100	400–580	140	52	6	115	A
21	2, 0, 0	30–80	300–580	141	62		16	A
	3, 0, 0	30–80		351	66	28	20	C
22	4, 0, 4	30–60	500–590	129	57	9	107	A
23	10, 0, 0	20–80	300–680	138	45	4	125	A
24	8, 0, 0	15–80	400–680	132	48	6	76	A
25	11, 0, 0	10–80	450–680	141	45	4	143	A
26	9, 0, 1	10–100	0–680	142	47	6	92	A
27	4, 0, 0	20–120	300–590	133	61	8	123	A
	19[b]	Mean A		134.6	54.1	3.5	93.6	A
	3[b]	Mean B		322.7	−33.9	23.1	29.6	B
	7[b]	Mean C		355.8	86.6	16.9	13.7	C

NOTE: The mean direction is given by its declination (D), inclination (I), radius of cone of 95% confidence (α_{95}), and precision parameter (k) (Fisher 1953). The M column lists the defined magnetization component.

[a]Number of specimens giving used end points (e), used remagnitization circles (r), and not used (o).

[b]Number of site means used for the component mean direction.

450–500°C on thermal step demagnetization, above which the A component is defined in the magnetite unblocking temperature range from 500 to 585°C (Figs. 3*c*, 3*d*), in the hematite range 600–675°C (Fig. 3*e*), and in both magnetite and hematite (Fig. 3*f*). The rapid acquisition of IRM by the massive granite specimens shows that magnetite is their dominant remanence carrier, and subsequent AF demagnetization of their IRM indicates that these carriers are mostly in the pseudosingle domain (PSD) to multidomain (MD) size range (Fig. 4). The characteristic A magnetization directions from individual sites are well grouped, with radii for their cones of 95% confidence (α_{95}) of <12° (Table 1). The site mean directions are also well clustered (Fig. 5*b*), with a unit mean of D = 134.6° (declination), I = 54.1° (inclination) (α_{95} = 3.5°, k = 93.6 (precision parameter), N = 19 (number of sites)).

The three foliated mafic dike sites all retain the B component. These specimens all start with a weak NRM intensity in the mid 10^{-3} A/m range. On AF step demagnetization, their NRM undergoes linear decay up to 100 mT after removal of initial VRM by 20 mT (Fig. 6*a*). Thermal demagnetization shows that almost all of the B component is removed between the 300 and 400°C step (Fig. 6*b*). This AF and thermal demagnetization behaviour is characteristic of pyrrhotite. Type curves for the IRM acquisition and demagnetization of pyrrhotite are not available for comparison. The specimens from sites 17 and 19 show a slower rate of IRM acquisition than for magnetite but faster than for hematite, and the AF demagnetization of the IRM shows a near-linear intensity decay rate that is atypical of either magnetite or hematite. The B characteristic directions for each site are moderately well grouped with k values of >30. The unit mean direction determined from the three dike sites is D = 323°, I = −34° (α_{95} = 23°, k = 30, N = 3).

Sites 1–3 are close to the Parker Lake shear zone and strongly sheared, and they retain the C component. Site 4 is more remote from the shear zone and retains both A and C components. C is also found at sites 15 and 16 that have a readily observable foliation and are close to an apparent linament followed by the Wathaman River. At site 21 both A and C magnetization components are present. This site is in foliated granite next to the mafic dikes. The C component specimens have NRM intensities that are in the mid 10^{-2} A/m

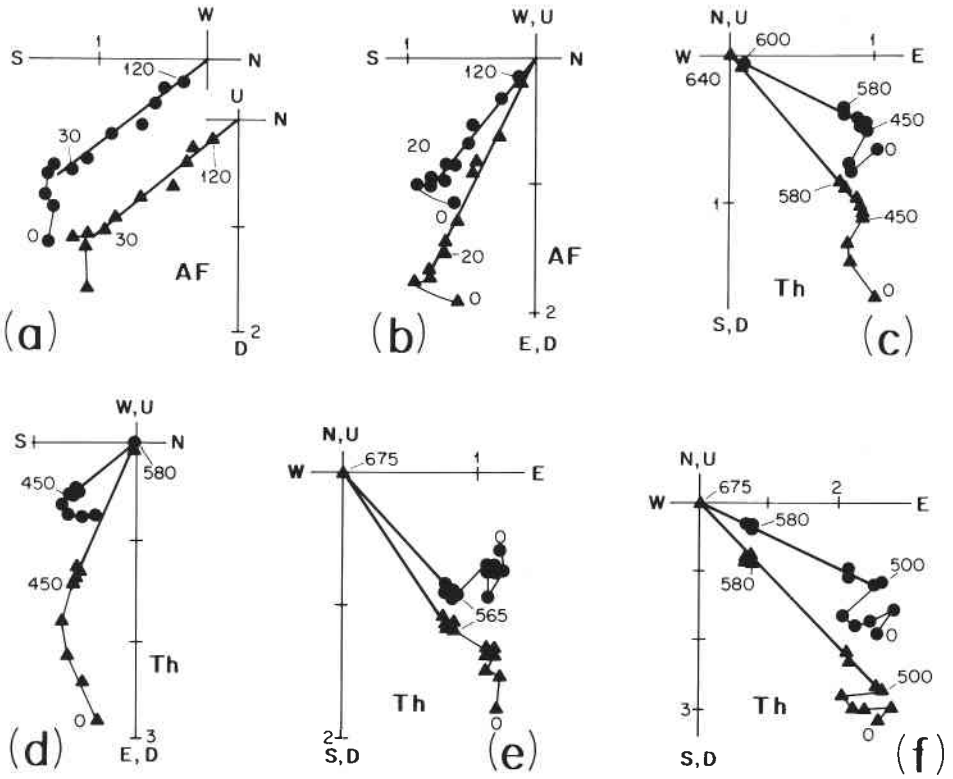


FIG. 3. Orthogonal plots of the decay of the remanence vector on alternating-field (AF) and thermal (Th) step demagnetization for example A component specimens from (*a*) site 10, (*b*) site 26, (*c*) site 11, (*d*) site 20, (*e*) site 23, and (*f*) site 9. Points on the horizontal (N, E, S, W) and vertical (U, D) planes are denoted by circles and triangles, respectively. The axial intensities are in 10^{-1} A/m, AF step field intensities in mT, and thermal step temperatures in °C.

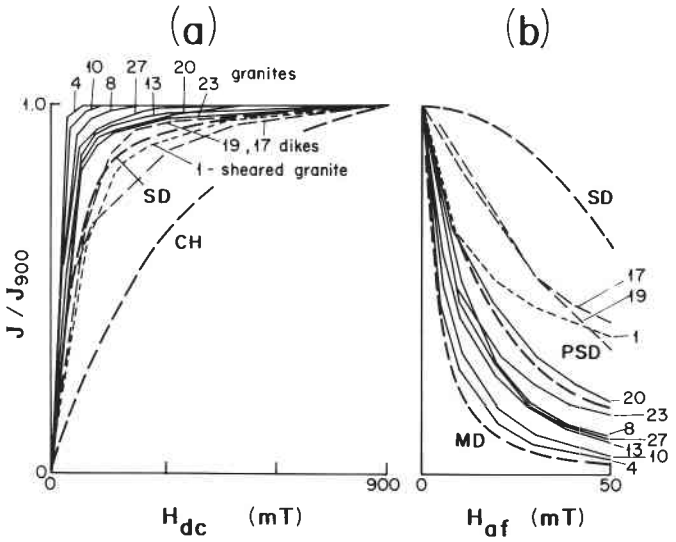


FIG. 4. Isothermal remanent magnetization (IRM) test specimen results, showing (*a*) acquisition of IRM and (*b*) AF demagnetization of IRM, relative to type curves for single domain (SD), pseudosingle domain (PSD), and multidomain (MD) magnetite and for coarse-grained hematite (CH). J/J_{900} is the ratio of the measured remanence intensity to IRM intensity at 900 mT, and H_{dc} and H_{af} are the applied magnetizing and demagnetizing fields, respectively.

range. AF step demagnetization shows, as before, the initial removal of VRM before isolation of the C magnetization (Fig. 6*c*). Thermal step demagnetization similarly shows the isolation of C in magnetite, hematite, and both magnetite and hematite unblocking temperature ranges (Figs. 6*d*, 6*e*). The IRM acquisition curve for a sheared granite specimen shows the rapid initial rise below 100 mT from magnetite followed by a continuing rise from 300 to 900 mT from hematite, and the rapid decay followed by notable flattening on AF demagnetization of the IRM confirms the presence of both magnetite and hematite as the C remanence carrier (Fig. 4). The C characteristic directions are well grouped within each site, with α_{95} values of 13° or less except for the two two-component sites (4 and 21) with reduced specimen numbers. The site means, however, form a relatively loose cluster with a unit mean remanence direction of $D = 356°$, $I = 87°$ ($\alpha_{95} = 17°$, $k = 14$, $N = 7$) (Fig. 5*a*). The large between-site scatter may reflect differential amounts of tectonic rotation and alteration between the sites on shearing.

Remanence stability tests

Three tests were attempted to assure that the A magnetization component in the batholith is primary.

First, three mafic pyroxenite dikes were sampled, at sites 17, 18, and 19. These dikes strike at 45 ± 5° NE and dip at 75 ± 10° NW. They are 0.56, 0.90, and 0.45 m wide, respectively, and are 0.30 and 0.51 m apart. Their B magnetization component is reversed in polarity, whereas the A component found throughout most of the batholith is normal. Therefore, the A and B components are not coeval. They could, however, be recording a reversal of the Earth's magnetic field within a time period of less than about one million years. When the B mean direction is switched to its antiparallel normal equivalent and compared with the A mean using the test of McFadden and Lowes (1981), the two populations are found to have comparable distributions but to be significantly different in direction at well over the 99% confidence level. Specimens from the baked contact of one dike were measured but failed

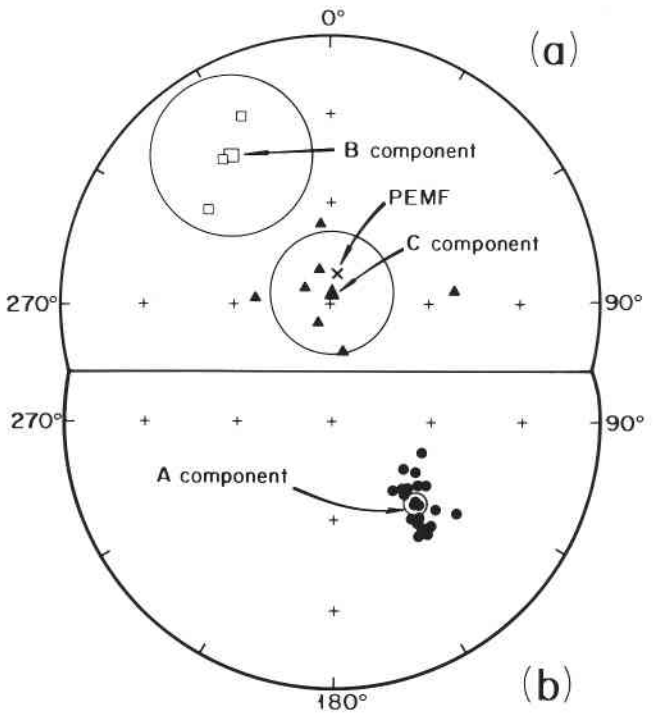


FIG. 5. Equal-area stereogram showing the site mean directions (small symbols) and unit mean directions (large symbols with their cone of 95% confidence) for (*a*) the B and C magnetization components and (*b*) the A component. Filled (open) symbols are plotted in the lower (upper) hemisphere. PEMF, present Earth's magnetic field direction.

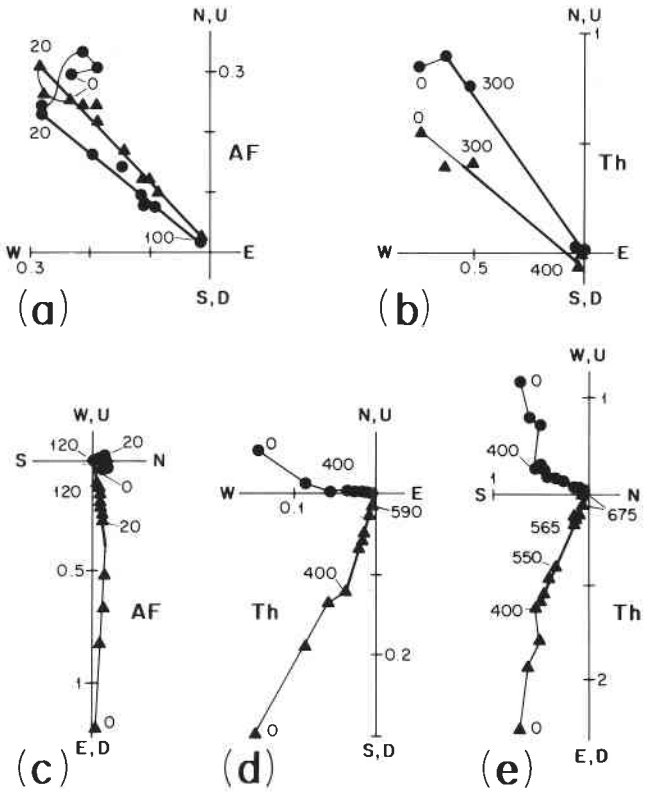


FIG. 6. Orthogonal plots of example specimens having the B magnetization component, (*a*) site 17 and (*b*) site 18, and the C component, (*c*) site 2, (*d*) site 1, and (*e*) site 15. Conventions as in Fig. 3.

to retain a stable remanent magnetization, so the paleomagnetic contact test (Everitt and Clegg 1962) was not executed. Specimens from sites 20 and 21 in the same outcrop but several metres from the dikes retain the A component. Obviously, however, the A and B components represent a time difference of several million years at least.

In the second test, sites were collected along a profile tracking southeast along Highway 105 from the Parker Lake shear zone (Fig. 2). Sites 1, 2, and 3 are in strongly foliated granite from close to the shear zone and retain the C magnetization component; site 4 retains both A and C components; and site 5 and onwards in massive granite retains the A component only (Table 1). In addition, the C component is found in foliated granites from sites 15 and 16 and in some specimens from site 20 next to the dikes. Noting that the cones of 95% confidence around the A and C mean directions do not intersect, it is evident that A and C represent significantly different magnetization ages. The C direction encompasses the present Earth's magnetic field direction of $D = 18°$, $I = 81°$ within its cone of 95% confidence. This suggests C could be a modern magnetization, but this is not likely because C is retained in both magnetite and hematite. In a few specimens from sites 1, 2, 4, and 15, there is evidence that C has not entirely replaced A as the latter appears to be present in the highest coercivity and unblocking temperature domains (Fig. 7*a*). There are also some dike specimens that retain the B magnetization in the higher coercivity and unblocking temperature domains, with B overprinted by the C component in the lower domains, (Fig. 7*b*). This evidence indicates that the C component is younger than both A and B.

A companion study is currently in progress on the paleomagnetism of the gabbros of the Peter Lake domain and their

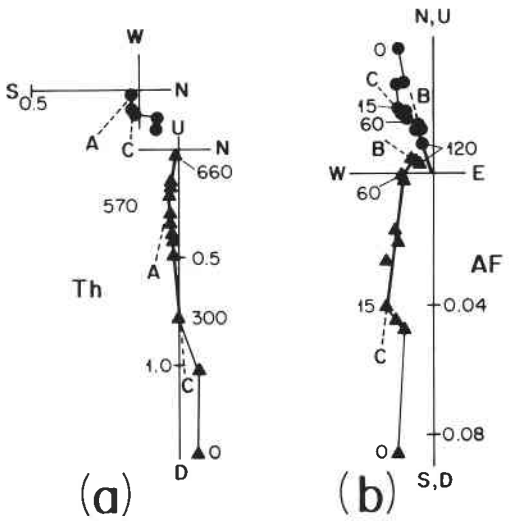


FIG. 7. Orthogonal plots of example specimens having (*a*) the C magnetization component overprinting the A component from site 2 and (*b*) the C component overprinting the B component from site 19. Conventions as in Fig. 3.

metamorphosed equivalents, the mafic Parker Lake gneisses, on the northern side of the study area. Preliminary data show the presence of the C magnetization component in the gneisses and a shallowly upward westward component in the gabbros. Since the Wathaman granite intrudes the Archean Peter Lake gabbros, these data provide a partial contact test that indicates that the younger A component in the batholith is primary.

Taken together the combination of geologic and magnetic evidence suggests that the A component in the batholith is

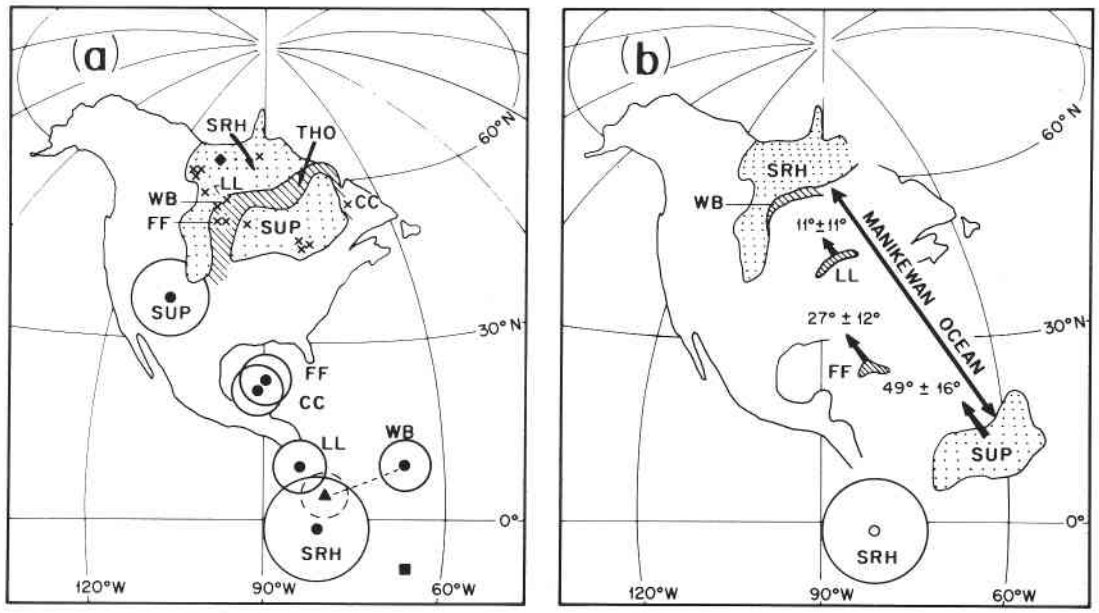


FIG. 8. Pole positions with their cone of circular standard deviation for 1825–1900 Ma rock units (×) in the Superior (SUP) and Slave-Rae-Hearne (SRH) provinces and Trans-Hudson orogen (THO) in (*a*) present position and (*b*) restored 1865 Ma position excluding the CC pole and terrane. The triangle shows the WB pole (A component) after correction in paleodeclination for a 15° counterclockwise rotation about a vertical axis. The locations of the poles for the B and C components in the dikes at 68°W, 8°S and granites at 105°W, 64°N are shown by a square and a diamond, respectively. CC, Castignon complex; FF, Flin Flon domain; LL, Lynn Lake – LaRonge domain; WB, Wathaman batholith.

primary, the B component in the dikes is younger, and the C component in the sheared granite is youngest.

Discussion of pole positions

There is no way, such as the distribution of petrologic phases, of determining paleohorizontal in the Wathaman batholith at present. Thus there is no way of knowing what tilt correction, if any, should be applied to the A unit mean direction. Therefore the initial assumption is that either there were no localized deformations or a large enough area was sampled to average them out, so the A magnetization component reflects only large-scale geotectonic motion. The small α_{95} value of 3.5° suggests that this assumption is valid. The pole position calculated for the A component is 67°W, 9°N (dp = 3°, dm = 5°) (Fig. 8*a*). This position is an excellent fit for an 1854 Ma age in paleoinclination, noting initially that it lies between the composite pole for the Slave–Rae–Hearne (SRH) craton on the batholith's northwestern side and the pole for the untilted vertical gabbro pipes in the Lynn Lake – LaRonge domain on its immediate southeastern side. The discordance in paleodeclination of the Wathaman pole is relatively minor and is discussed later.

The pole derived for the dikes' B component is located at 68°W, 8°S (dp = 15°, dm = 26°). This pole is too poorly defined to provide more than an indication of the age of the dikes' remanence. Since the B pole with its large oval of confidence is beside the A pole, and the SRH pole, the indication is that the B acquisition age is no more than about 25 Ma younger than the A age, assuming a typical rate of apparent polar wander (APW) of about 0.5° per million years. If the dikes were vertical when emplaced, the corrected pole would be shifted a little southward to about 65°W 16°S, thereby degrading its fit to both the batholith's and SRH poles. This suggests but does not prove that the dikes were emplaced along weakened foliation planes and acquired their remanence when pyrrhotite was crystallized during the dike's alteration to pyroxenite. Noting that pyrrhotite has much lower unblocking temperatures (300–350°C) than magnetite and hematite (>500°C), another possibility is that the B magnetization is recording a slightly later time in the cooling history of the batholith than the A component.

The pole obtained for the C remanence in the sheared granites is located at 105°W, 64°N (dp = 33°, dm = 34°). Like the B pole, the C pole has such large error limits that it provides only a crude indication of C's acquisition age. One source of scatter could be that the C sites are located in discrete blocks, so their mean remanence directions have been rotated with the blocks by the shearing. The pole itself fits on the APW path of Irving (1979) at about 1650 Ma which is not sensible given the older limit of 1854 ± 11 Ma provided by the U/Pb zircon dates on the batholith and the younger limit of 1735 ± 15 Ma provided by the onset of deposition of the Athabaska Formation. The probable explanation is that this segment of the Irving (1979) APW path for the North American craton is not very reliable because it is derived mostly from Greenland poles that have an uncertain tectonic correction relative to the western Canadian Shield (Hoffman 1989).

Trans-Hudson closure

The paleomagnetic pole from the Wathaman batholith is a useful reference point for discussing the closure of the Manikewan Ocean to form the THO. Dunsmore and Symons (1990) discussed at length the limitations of the paleomagnetic and radiometric data for the 1825–1890 Ma poles of the Canadian Shield that are summarized in Table 2. The poles for the Slave, Rae, and Hearne provinces form one coherent cluster, with a mean pole (SRH, Fig. 8*a*) that is distinct from the mean of the Superior Province cluster of poles (SUP); the THO poles fall between them. Further, this array of poles lies more or less on a great circle that is fairly close to, and about parallels, a great

TABLE 2. Pole positions for the Trans-Hudson orogen, 1825–1890 Ma

Pole	Units[a]	Long. (°W)	Lat. (°N)	CSD (°)
SRH	Slave, Rae, and Hearne provinces, Trans-Hudson orogen	82	−2	8
WB	Wathaman batholith	67	9	3[b]
LL	Lynn Lake – LaRonge domain	84	9	4[b]
FF	Flin Flon domain	90	24	4[b]
CC	Castignon complex	92	22	8[b]
SUP	Superior Province	110	38	8

[a]A complete discussion of the paleomagnetic-pole data and assigned ages is given in Dunsmore and Symons (1990) and is simply summarized here. SRH poles: Martin Formation (1835 Ma), Dubawnt Group (1825 Ma), Kohochella Group (1850 Ma), Seton Formation (1870 Ma), Akaitcho Formation (1890 Ma); WB pole: Wathaman batholith (1854 Ma); LL pole: Lynn Lake gabbros (1890 Ma); FF poles: Amisk Group (1855 Ma), Boundary intrusions (1855 Ma); CC pole: Castignon complex (1880 Ma); SUP poles: Molson dikes (1884 Ma), Spanish River compex (1838 Ma), Sudbury norite (1850 Ma), Sudbury granophyre (1849 Ma).

[b]Value of circular standard deviation (CSD) (Fisher 1953) is based on site means rather than pole means.

circle through the paleomagnetic study areas, so the significant difference is in the paleoinclination of the pole groups. Specifically the poles record a progressive decrease in paleoinclination from high paleolatitudes in the Superior Province to low paleolatitudes in the SRH provinces. This is fortunate because it means that terrane rotations can only be a minor cause of discordance in the poles and that translation must be the main cause.

The Proterozoic Wathaman batholith's pole (WB) agrees perfectly with the SRH pole in terms of paleoinclination (Fig. 8*a*). This confirms that the batholith represents a plutonic welt formed along the margin of the Archean Hearne province over a suture zone and was not rafted into place. It also confirms that the batholith has not undergone regional tilt about a horizontal axis that is parallel to the northeasterly trend of the batholith in the study area, as this would change the paleoinclination of the batholith's pole. The discordance in paleodeclination of the WB pole can be best explained by one of two mechanisms. One is that the batholithic arc was originally a little straighter and that its southwestern arm has been rotated counterclockwise by about 15° in the orogenic collision as it impacted on a relatively rigid cratonic margin. The alternative mechanism is that the batholith has undergone about 17° or 1800 km of right-lateral transcurrent motion along the Needle Falls and Parker Lake shear zones. Both alternatives are consistent with the kinematics of existing geotectonic models (e.g., Hoffman 1988, 1990), although the 17° magnitude for the transcurrent model seems unreasonably large, given that the shear zone is only about 700 km long at present. It is also possible that both mechanisms contributed in part to the paleodeclination discordance.

Holding the SRH craton and Wathaman batholith fixed, the Lynn Lake – LaRonge domain, Flin Flon domain, and Superior Province have to be backed up to the southeast by about 11 ± 11°, 27 ± 12°, and 49 ± 16°, respectively, to bring the poles into concordance (Fig. 8*b*), and this indicates that these terranes were translated northwestward by at least these amounts during closure of the THO. This result indicates that the Manikewan Ocean was in the order of 5000 km wide at about 1865 Ma, a little wider than the original estimate of Dunsmore and Symons (1990) or about equivalent to the present-day Atlantic Ocean. It also fits with the abundant evidence in the THO for extreme crustal shortening (Lewry *et al.* 1990).

The separation of 16 ± 8° in paleolatitude between the Lynn Lake – LaRonge and Flin Flon domains indicates that these volcanic intraoceanic arcs (Gordon *et al.* 1990) were separated by about 1700 km of oceanic basin at formation. This suggests that they are not faulted or offset segments of the same volcanic belt as proposed by Green *et al.* (1985*a*, 1985*b*). Noting that the assigned magnetization ages for the two domains differ by 35 Ma, this result is still very speculative. Work underway in the THO should help determine the difference more precisely.

The Castignon Lake carbonatite complex in the New Quebec segment of the THO has been more precisely dated by U/Pb zircon at 1880 ± 2 Ma by Chevé and Machado (1988) than is shown by the age listed in Dunsmore and Symons (1990). Its geotectonic movement is less certain, as its pole position (CC, Fig. 8*a*) could be readily explained by northwesterly translation, by clockwise rotation, or by a combination of these motions for the New Quebec segment. More paleomagnetic data are needed from this segment to constrain the geotectonic options.

The fundamental determination of relative northwestward translation of the THO domains and Superior craton into the suture with northwest-dipping subduction under the Wathaman batholith is entirely consistent with the structural evidence and geotectonic models for Andean-type convergence and collision of the Hearne and Superior provinces to form the THO in the western Canadian Shield (Gibb *et al.* 1980, Lewry and Sibbald 1980; Ray and Wanless 1980; Lewry 1981; Lewry *et al.* 1981, 1990; Gibb 1983; Stauffer 1984; Green *et al.* 1985*a*, 1985*b*; Thomas and Gibb 1985; Hoffman 1987, 1988, 1989, 1990; Van Schmus *et al.* 1987; Gordon *et al.* 1990; Halden *et al.* 1990). As additional paleomagnetic data become available for other domains and for a range of times within each domain of the THO, it should be possible to closely constrain the limits of geotectonic motion of each terrane much more closely both in magnitude and direction.

Acknowledgments

I wish to thank Pan Hairu, who assisted in the collecting, L. McCrone, who prepared the samples, J. Kulhanek, who did the remanence measurements, and the Natural Sciences and Engineering Research Council of Canada, who funded this work. I'm also indebted to D. J. Dunlop and M. R. Stauffer for many useful comments to improve the manuscript.

BAILEY, R. C., and HALLS, H. C. 1984. Estimate of confidence in paleomagnetic directions derived from mixed remagnetization circle and direct observational data. Journal of Geophysics, **54**: 174–182.

CHEVE, S., and MACHADO, N. 1988. Reinvestigation of the Castignon Lake carbonatite complex, Labrador Trough, New Quebec. Geological Association of Canada, Program with Abstracts, **8**: A20.

DUNSMORE, D. J., and SYMONS, D. T. A. 1990. Paleomagnetism of the Lynn Lake gabbros in the Trans-Hudson Orogen and closure of the Superior and Slave cratons. *In* The Early Proterozoic Trans-Hudson Orogen of North America. *Edited by* J. F. Lewry and M. R. Stauffer. Geological Association of Canada, Special Paper 37, pp. 215–228.

EVERITT, C. W. F., and CLEGG, J. A. 1962. A field test of paleomagnetic stability. Geophysical Journal of the Royal Astronomical Society, **6**: 312–319.

FISHER, R. A. 1953. Dispersion on a sphere. Proceedings of the Royal Society of London, Series A, **217**: 295–305.

FUMERTON, S. L., STAUFFER, S. L., and LEWRY, J. F. 1984. The Wathaman batholith: largest known Precambrian pluton. Canadian Journal of Earth Sciences, **21**: 1082–1097.

GIBB, R. 1983. Model for suturing of Superior and Churchill plates: An example of double identation tectonics. Geology, **11**: 413–417.

GIBB, R. A., THOMAS, M. D., and MURKHOPADHYAY, M. 1980. Proterozoic sutures in Canada. Geoscience Canada, **7**: 149–154.

GORDON, T. M., HUNT, P. A., BAILES, A. H., and SYME, E. C. 1990. U–Pb ages from the Flin Flon and Kisseynew belts, Manitoba: Chronology of crust formation at an Early Proterozoic accretionary margin. *In* The Early Proterozoic Trans-Hudson Orogen of North America. *Edited by* J. F. Lewry and M. R. Stauffer. Geological Association of Canada, Special Paper 37, pp. 177–199.

GREEN, A. G., HAJNAL, Z. and WEBER, W. 1985*a*. An evolutionary model of the western Churchill Province and western margin of the Superior Province in Canada and the north-central United States. Tectonophysics, **16**: 281–322.

GREEN, A. G., WEBER, W., and HAJNAL, Z. 1985*b*. Evolution of Proterozoic terrains beneath the Williston Basin. Geology, **13**: 624–628.

HALDEN, N. M., CLARK, G. S., CORKERY, M. T., LENTON, P. G., and SCHLESWITZ, D. C. P. 1990. Trace-element and Rb–Sr whole-rock isotopic constraints on the origin of the Chipewyan, Thorsteinson, and Baldock batholiths, Churchill Province, Manitoba. *In* The Early Proterozoic Trans-Hudson Orogen of North America. *Edited by* J. F. Lewry and M. R. Stauffer. Geological Association of Canada, Special Paper 37, pp. 201–214.

HOFFMAN, P. F. 1987. Continental transform tectonics: Great Slave Lake shear zone (1.9 (Ga), northwest Canada. Geology, **15**: 785–788.

——— 1988. United plates of America, the birth of a craton; Early Proterozoic assembly and growth of Laurentia. Annual Review of Earth and Planetary Sciences, **16**: 543–603.

——— 1989. Precambrian geology and tectonic history of North America. *In* Geology of North America—An overview. *Edited by* A. W. Bally and A. R. Palmer. Geological Society of America, Geology of North America, Vol. A, pp. 447–512.

——— 1990. Subdivision of the Churchill Province and extent of the Trans-Hudson Orogen. *In* The Early Proterozoic Trans-Hudson Orogen of North America. *Edited by* J. F. Lewry and M. R. Stauffer. Geological Association of Canada, Special Paper 37, pp. 15–39.

IRVING, E. 1979. Paleopoles and paleolatitudes of North America and speculations about displaced terrains. Canadian Journal of Earth Sciences, **16**: 669–694.

JOHNSTON, W. Q. 1985. Geology of the Wathaman Lake (East) area. Saskatchewan Energy and Mines, Open File Report 85–1.

KIRSCHVINK, J. L. 1980. The least-squares line and plane and the analysis of paleomagnetic data. Geophysical Journal of the Royal Astronomical Society, **62**: 699–718.

LEWRY, J. F. 1981. Lower Proterozoic arc–microcontinent collisional tectonics in the western Churchill Province. Nature (London), **294**: 69–72.

LEWRY, J. F., and SIBBALD, T. I. I. 1980. Thermotectonic evolution of the Churchill Province in northern Saskatchewan. Tectonophysics, **68**: 45–82.

LEWRY, J. F., CURRY, A., and ROBERTS, K. 1980*a*. Geology of an area around Ghana Lake. Saskatchewan Energy and Mines, Report 192.

LEWRY, J. F., ROBERTS, K., and REES, C. J. 1980*b*. Geology of the area around Spalding Lake. Saskatchewan Energy and Mines, Report 199.

LEWRY, J. F., STAUFFER, M. R., and FUMERTON, S. L. 1981. A Cordilleran-type batholithic belt in the Churchill Province in northern Saskatchewan. Precambrian Research, **14**: 277–313.

LEWRY, J. F., THOMAS, D. J., MACDONALD, R., and CHIARENZELLI, J. 1990. Structural relations in accreted terranes of the Trans-Hudson Orogen, Saskatchewan, telescoping in a collisional regime? *In* The Early Proterozoic Trans-Hudson Orogen of North America. *Edited by* J. R. Lewry and M. R. Stauffer. Geological Association of Canada, Special Paper 37, pp. 75–94.

MACHADO, N. 1990. Timing of collisional events in the Trans-Hudson Orogen: evidence from U–Pb geochronology for the New Quebec Orogen, the Thompson Belt, and the Reindeer Zone (Manitoba and Saskatchewan). *In* The Early Proterozoic Trans-Hudson Orogen of North America. *Edited by* J. F. Lewry and M. R. Stauffer. Geological Association of Canada, Special Paper 37, pp. 433–441.

MCFADDEN. P. L., and LOWES, F. J. 1981. The discrimination of mean directions drawn from Fisher distributions. Geophysical Journal of the Royal Astronomical Society **67**: 19–33.

MEYER, M. T. 1987. Geochronology and geochemistry of the Wathaman Batholith, the remnant of an Early Proterozoic continental arc in the Trans-Hudson orogen, Saskatchewan, Canada. M.Sc. thesis, University of Kansas, Lawrence, KS.

RAY, G. E. and WANLESS, R. D. 1980. The age and geological history of the Wollaston, Peter Lake, and Rottenstone domains in northern Saskatchewan. Canadian Journal of Earth Sciences, **17**: 333–347.

STAUFFER, M. R. 1984. Manikewan: an early Proterozoic ocean in central Canada, its igneous history and orogenic closure. Precambrian Research, **25**: 257–281.

——— 1990. The Missi Formation: an Aphebian molasse deposit in the Reindeer Lake Zone of the Trans-Hudson Orogen, Canada. *In* The Early Proterozoic Trans-Hudson Orogen of North America. *Edited by* J. F. Lewry and M. R. Stauffer. Geological Association of Canada, Special Paper 37, pp. 121–141.

THOM, A., ARNDT, N. T., CHAUVEL, C., and STAUFFER, M. R. 1990. Flin Flon and Western LaRonge belts, Saskatchewan: products of Proterozoic subduction-related volcanism. *In* The Early Proterozoic Trans-Hudson Orogen of North America. *Edited by* J. F. Lewry and M. R. Stauffer. Geological Association of Canada, Special Paper 37, pp. 163–175.

THOMAS, M. D., and GIBB, R. A. 1985. Proterozoic plate subduction and collision: Processes for reactivation of Archean crust in the Churchill Province. *In* Evolution of Archean supracrustal sequences. *Edited by* L. D. Ayres, P. C. Thurston, K. D. Card, and W. Weber. Geological Association of Canada, Special Paper 28, pp. 263–279.

VAN SCHMUS, W. R., BICKFORD, M. E., LEWRY, J. F., and MACDONALD, R. 1987. U–Pb geochronology in the Trans-Hudson Orogen, Canada. Canadian Journal of Earth Sciences, **24**: 407–424.